

# Optimality Theory

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## 1 Introduction

Optimality Theory (OT; Prince and Smolensky, 1993/2004, 1997; McCarthy and Prince, 1993b) is a constraint-based framework for modeling input-output mappings. While it has primarily been used as a model of phonology, it has been used in other linguistic domains, especially syntax (Legendre et al., 1993; Grimshaw, 1997; Barbosa et al., 1998; Legendre et al., 2001; Broekhuis and Vogel, 2013; Murphy, 2026), and has even been extended beyond linguistics (Dresher, 1996; Parker and Parker, 2004; Biró, 2011). The basic idea behind OT is that attested input-output mappings outperform alternative, logically possible mappings according to a set of constraints. Exactly how this works depends on the framework’s three components, GEN, CON, and EVAL.

The goal of this chapter is to introduce these components, building a picture of the framework from the ground up. Due to its length, the chapter is necessarily dense, but I have taken care to illustrate the abstract and formal mechanisms with concrete examples. There are several book-length introductions to OT, including Prince and Smolensky (1993/2004), Archangeli and Langendoen (1997), Kager (1999), McCarthy (2002b, 2008a), and Tessier (2015). For discussion of its history, see Archangeli (1997), Burzio (2000), Fikkert and de Hoop (2009), Scheer (2010), Uffman (2010), Griffiths (2019), and van Oostendorp (2022).

The following three sections of the chapter correspond to the three components of OT. Section 2 discusses where the competing forms come from, section 3 discusses how they are evaluated by the grammar, and section 4 discusses how the grammar picks one as the output. Section 5 summarizes.

## 2 The competitors: GEN

Unlike rule-based frameworks such as Chomsky and Halle (1968), Optimality Theory models input-output mappings as competition between different logically possible outputs. Abstractly, to model an observed mapping  $/A/ \rightarrow [B]$ , it must be the case that  $[B]$  is better than all other possible outputs ( $[C]$ ,  $[D]$ , ...) including  $[A]$  itself. This section discusses where these competitors come from; the following two sections discuss the notion of one being *better than* another.

A common way to model allomorphy is by storing different forms in the lexicon (Mascaró, 1996; Paster, 2006; Nevins, 2011; Bonet et al., 2015; Smith, 2015; Bonet, 2023). As a concrete example, Lee (2008) analyzes the nominative suffix in Korean (**kor**; Koreanic) along these lines; it has two forms, [-i] and [-ga], with the former surfacing after consonants (1a-c) and the latter after vowels (1d-f). For any given noun, there are (at least) two logically possible output forms that compete, such as [sok.k'a.] vs. [so.gi.] and [ha.ma.ga.] vs. [ha.ma.i.]. In an OT analysis, the attested surface forms [so.gi.] and [ha.ma.ga.] must outperform their competitors \*[sok.k'a] and \*[ha.ma.i.], respectively.

(1) Korean nominative (a-d: Lee, 2008:69; e-f: Hanbyul Song, p.c.)

- |    |              |                 |                    |
|----|--------------|-----------------|--------------------|
| a. | [.sok.]      | [.so.gi.]       | ‘inside-NOM’       |
| b. | [.waŋ.]      | [.waŋ.i.]       | ‘king-NOM’         |
| c. | [.mul.]      | [.mu.ri.]       | ‘water-NOM’        |
| d. | [.so.]       | [.so.ga.]       | ‘cow-NOM’          |
| e. | [.ha.ma.]    | [.ha.ma.ga.]    | ‘hippopotamus-NOM’ |
| f. | [.ho.raŋ.i.] | [.ho.raŋ.i.ga.] | ‘tiger-NOM’        |

Not all competing forms are plausibly listed in the lexicon, and must instead be generated by the grammar. For example, several verbal suffixes in Korean surface with an initial [i] after consonants (2a,c) and without it after vowels (2b,d). While listing these forms lexically would be empirically adequate, it would be uneconomical and would fail to capture the generalization that [i] does not surface in vowel chains elsewhere in the language, as in (3). Further, if speakers stored all possible forms, we would not expect them to extend phonological generalizations to novel items, which they reliably do (Berko, 1958; Jun, 2010; Jun and Albright, 2017).

(2) Korean purposive and conditional (Hong, 2001:216)

- |    |         |                |            |
|----|---------|----------------|------------|
| a. | [.mæk.] | [.mæ.gi.rə.]   | ‘eat-PURP’ |
| b. | [.sa.]  | [.sa.rə.]      | ‘buy-PURP’ |
| c. | [.mæk.] | [.mæ.gi.mjən.] | ‘eat-COND’ |
| d. | [.sa.]  | [.sa.mjən.]    | ‘buy-COND’ |

(3) Korean [i] ~ [Ø] alternations (a,c: Hanbyul Song, p.c.; b,d: Kim-Renaud, 2022:42-43)

- |    |             |               |                    |
|----|-------------|---------------|--------------------|
| a. | /s'i-ta/    | [.s'i.da.]    | ‘to use’           |
| b. | /s'i-əto/   | [.s'ə.do.]    | ‘even if one uses’ |
| c. | /camki-ta/  | [.cam.gi.da.] | ‘to immerse’       |
| d. | /camki-ita/ | [.cam.gi.da.] | ‘is immersed’      |

In OT, the generator function GEN is responsible for producing a set of competing forms or *candidates* from an input. As an illustration, the set of candidates generated from /sok/

includes the *fully faithful candidate* [.sok.], candidates with different feature specifications like [.θok.] and the extremely divergent [.Ōũg.], candidates with fewer segments such as [.so.] and [.o.], candidates with more segments such as [.psok.] and [.so.kə.], and all other logically possible candidates. While the preceding examples are all syllabified, prosodic structure is another dimension of variation within the candidate set.

Unlike rules, which may be language-specific, Prince and Smolensky (1993/2004) propose that GEN is universal; see McCarthy (1993) and Blevins (1997) for alternative approaches. Accordingly, because alternations between clicks and non-clicks are attested (see Bennett, 2020 for an overview), [.Ōũg.] is in the candidate set of /sok/ despite Korean lacking clicks. Their absence in the language derives from their losing to candidates without clicks rather than their logical impossibility. This example also raises the question of modality: does GEN produce [.Ōũg.] as a candidate from a signed input? See Brentari (1999) and Eccarius (2008) for discussion of sign languages in OT.

Another property of GEN, commonly referred to as *freedom of analysis* (e.g., McCarthy, 2002b:8), is its producing candidates that are arbitrarily different from the input. This property derives infinitely large candidate sets: if a candidate is generated by inserting  $n$  segments, then there must also be a candidate generated by inserting  $n + 1$  segments, meaning that there is no longest candidate. One consequence is that the candidate set generated from /sok/ contains [.so.kə.], [.so.kə.ə.], [.so.kə.ə.ə.], and so on, and is densely populated with what Prince and Smolensky (1993/2004:227) dub *supererogatory garbage* (cited page numbers refer to the 2004 monograph). Usually, as the next section discusses, if any candidate with a large number of inserted segments loses the competition, so too should those with even more inserted segments. As Lamont (2024a) argues, however, there are instances where this filtering fails.

Allowing the grammar access to infinitely many candidates is yet another reason not to list them all out but instead to use the GEN function as a finite description of an infinite set. That a component like this should be universal and therefore innate is exactly inline with the tenets of generative linguistics (Chomsky, 1959), which posits that people acquire a finite description of language rather than parroting their finite experienced utterances. GEN is not the full description of a grammar, but it is a crucial starting point.

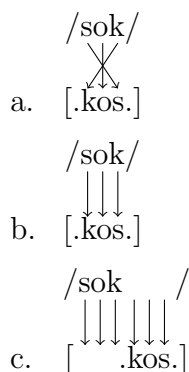
There have been proposals that limit how much epenthesis GEN can do (Łubowicz, 2003a; de Lacy, 2007; Golston, 2007), imposing a finite upper bound on the number of candidates produced from a given input (at least in terms of length; recursive prosodic structure – see e.g., Itô and Mester, 2021 – generates infinitely many candidates). However, while this avoids discomfort with infinities, the number of candidates is still absolutely enormous. For example, in order to model metathesis (see Hume, 1998, Heinz, 2005, Buckley, 2011, Canfield, 2015, and Chandlee, 2023 for discussion), GEN must be able to reorder segments. Freedom of analysis implies that a candidate set contains every permutation of the input; for example, from /sok/, GEN produces [.sko.], [.osk.], [.oks.], [.kso.], and [.kos.], among many other candidates. The number of permutations of an input with  $n$  segments is  $n! = n \times (n - 1) \times (n - 2) \times \dots \times 2 \times 1$ , which grows extremely quickly as  $n$  grows (operations that build structure also contribute very large numbers of candidates; see Karttunen, 1998, Prince, 2010, Bennett, 2015:29-30, and Bellik and Kalivoda, 2019). As a consequence, the number of candidates produced by GEN is extremely large and, even if finite, impractical to exhaustively enumerate (Ellison, 1994; Eisner, 1997a, 2000b; Idsardi, 2006; Lamont, 2019).

This idea of factorial growth will come up again in section 4.

## 2.1 Correspondence

Thus far, I have presented candidates simply as phonological forms; i.e., strings of segments augmented with prosodic structure such as syllables. This simplification creates undesirable ambiguity between superficially identical forms; as the following section discusses in more detail, it is necessary to keep track of how GEN produces each candidate. For example, while [.kos.] can be generated from /sok/ by rearranging its segments (4a), it can also be generated by changing the /s/ to [k] and the /k/ to [s] (4b) or by deleting all three segments and inserting three new segments (4c).

### (4) Ambiguity of generation

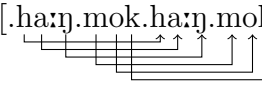
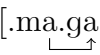
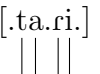


Keeping a ledger of mutations is the responsibility of *Correspondence Theory* (McCarthy and Prince, 1994, 1995, 1999), which is so widely assumed that it is practically synonymous with *Optimality Theory*. Under Correspondence Theory, candidates are augmented with their relation to the input. I represent this relation using underscript indices on segments; for example, from the input /sok/, [.kos.] represents (4a), [.kos.] represents (4b), and [.kos.] represents (4c).

Strictly speaking, under Correspondence Theory, it is not correct to describe a candidate [.so.] generated from /sok/ as having undergone deletion. Instead, one would say that the underlying /k/ has no *correspondent* in the candidate. To avoid using process-oriented terms like *epenthesis* and *deletion*, I adopt the more surface-oriented terms *presence* and *absence* (Werle, 2007).

Because this chapter's empirical focus is on input-output mappings, I only use the correspondence relation sketched above. It is, however, worth noting that correspondence has been generalized to a large number of domains. In reduplication, there is correspondence between copied segments (McCarthy and Prince, 1994, 1995, 1999) as in (5a). Vowel and consonant dis/harmony is often modeled as correspondence between interacting segments (Rose and Walker, 2004, 2011; Hansson, 2010; Bennett, 2015) as in (5b). Within paradigms, there may be correspondence between morphologically related forms (Benua, 1997; Alderete, 1999) as in (5c). There are also proposals establishing correspondence within segments (Shih and Inkelas, 2018). It is a rich representational tool.

(5) Other applications of correspondence relationships

- a. [ha:ŋ.mok.ha:ŋ.mok.] ‘an item by an item’ (Chung, 1999:45)  

- b. [ma.ga.] ‘block-DECL’ (Ko, 2022:193)  

- c. [ta.ri.]  
  
 [ta.ri.e.sə.] ‘bridge’, ‘bridge-LOC’ (Ko, 2006:223)

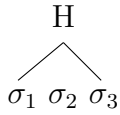
## 2.2 Restrictions on GEN

While Freedom of Analysis endows GEN with substantial power, it is not entirely unrestrained (see Blaho et al., 2007 for an overview). One standard assumption is that GEN cannot modify morphological affiliation, a property dubbed *Consistency of Exponence* (McCarthy and Prince, 1993b:21-22); see Łubowicz (2003a), Walker and Feng (2004), and Feng (2006) for alternative proposals and van Oostendorp (2007) for critical discussion. As an illustration, consider the input /sa-rə/ which surfaces as [sa.rə.] ‘buy-PURP’ in Korean. In all candidates generated from the input, all correspondents of the verb root /sa-/ retain the properties of being a verb, being a root, and belonging to the morpheme meaning ‘to buy’. Likewise, all correspondents of the suffix /-rə/ retain the properties of being a suffix and belonging to the purposive. Segments without correspondents in the input, such as epenthetic segments, have no morphological affiliation because GEN is unable to provide one.

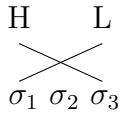
Other substantive restrictions, such as bounding the number of insertions as mentioned earlier, have been proposed on the operations available to GEN. Staroverov (2014) proposes that GEN is unable to insert consonants, and must instead fission them off adjacent vowels. Along similar lines, Prince and Smolensky (1993/2004:96-97) propose that GEN cannot delete segments in any sense: underlying segments that are absent on the surface are phonologically present, but uninterpretable by the phonetics. This restriction, *Containment Theory*, was partially superseded by Correspondence Theory, but has since reentered the literature (van Oostendorp, 2006, 2007; Revithiadou, 2007; Trommer, 2011; Zimmermann, 2017).

There have also been discussions of how GEN can manipulate autosegmental structures (Leben, 1973, 2011; Goldsmith, 1976), inheriting questions of well-formedness from the pre-OT literature. Ní Chiosáin and Padgett (2001) propose that GEN cannot create gapped structures, such as that in (6), while Kimper (2012) argues for their generation. As to whether GEN can create crossing autosegmental association lines, as in (7), while some authors assume it can (Akinlabi, 1996; Yip, 2002) and others assume it cannot (Pulleyblank, 1997; Beckman, 1998; Zimmermann, 2017), de Lacy (2007) argues that structures like these are removed by other components of the grammar. Restrictions on GEN raise empirical and analytical questions. If GEN cannot produce a given structure, then we expect it to be categorically unattested (so long as it is not present underlyingly). If GEN can produce an undesirable structure, then the grammar must ensure that those candidates fail to surface.

- (6) A gapped structure: the high tone (H) is associated to syllables 1 and 3, but not syllable 2

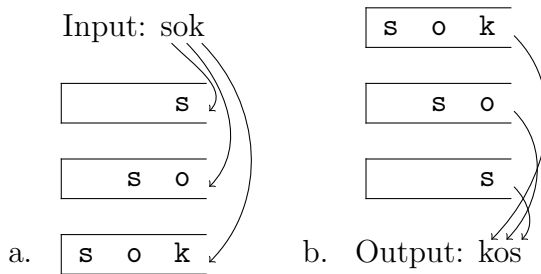


- (7) Crossing autosegmental lines: the high tone (H) is associated to syllable 3 and the low tone (L) is associated to syllable 1



Computational characterizations of GEN have argued that it should be finite-state (El-lison, 1994; Frank and Satta, 1998; Eisner, 1997a, 2000a, 2002; Riggle, 2004; Hulden, 2017); for formal introductions to finite-state machines see Chandlee (2014), Gorman and Sproat (2021), and [Chandlee and Jardine \(this volume\)](#). Informally, this means that it can only remember a fixed amount of information, which prevents it from modeling certain transformations such as reversals (Behrenfeldt, 2009; Wang and Hunter, 2023), e.g., *phonology*  $\rightarrow$  *ygolonohp*. In order to reverse an input, a model must be able to store the entire input in memory: each segment is stored in memory (8a) and written to the output only after the rest of the input is read and its corresponding output has been written (8b). Assuming that there is no upper bound on how large inputs can be, it is impossible for a model with a finite amount of memory to reverse all inputs.

- (8) String reversal



One reason to restrict GEN to a finite-state model is to limit its power to that of rewrite rules (Johnson, 1972; Koskeniemi, 1983; Kaplan and Kay, 1994; Chandlee and Jardine, 2021). It is impossible to write a rule or set of rules  $A \rightarrow B / C \_ D$  that reverses inputs. This appears to match the attested typology: not only are there no phonological processes that reverse inputs (Moreton et al., 2021), there do not appear to be any that are not finite-state (see Heinz, 2018 for an overview). Like other restrictions, a finite-state GEN raises empirical questions – among them being the treatment of metathesis. Only a strict subset of permutations are accessible by a finite-state GEN (reversals are a type of permutation), and this raises the question of how much reordering GEN can do. It must be possible to move segments more than one position, but a maximum distance has not been established. For example, in Mutsun (*css*; Yok-Utian; Okrand, 1977; Butler, 2013), the plural suffix alternates between [-mak] and [-kma], the former appearing after consonants (9a-b) and

the latter after vowels (9c-d). Following Butler’s (2013) analysis that the underlying form is /-mak/, the dorsal stop must move two positions to surface as [-kma]. The complexity of GEN may also depend on the domain of analysis, complicating questions of universality. Analyses of the syntax-phonology interface (e.g., Selkirk, 1995, 1996, 2011; Kusmer, 2020; Bellik et al., 2023) may need to manipulate trees, requiring non-finite state methods.

(9) Mutsun plural (Butler, 2013:191)

- a. [.weɪ.ren.mak.] ‘rabbit-PL’
- b. [.wim.mah.mak.] ‘wing-PL’
- c. [.lup.pek.ma.] ‘flat bulrush-PL’
- d. [.a.mak.ma.] ‘people; bodies-PL’

Exactly what GEN is and what it can do has largely taken a backseat to other theory-internal questions in OT. However, this is only the case in what is more correctly deemed *parallel Optimality Theory*, where, because GEN is unrestricted, it models multiple changes simultaneously, in parallel. In *Harmonic Serialism* (HS), GEN is restricted to producing candidates that differ from the input by applying at most one operation (Prince and Smolensky, 1993/2004; McCarthy, 2000, 2006, 2007b, 2016, [this volume](#); Pruitt, 2023); mappings that require multiple changes pass through a number of intermediate steps from input to output. This necessarily begs the question of what exactly GEN can do and is an active area of research in HS (McCarthy, 2009). For example, Takahashi (2019), Mooney (2022, 2023), and Lamont (2024b) answer the permutation question in the negative, arguing that GEN cannot rearrange phonological objects by applying a single operation.

### 3 Their compliances: CON

As the previous section laid out, the generator function GEN takes an input and produces a set of candidates, i.e., phonological forms and their relationship to the input. The grammar is now tasked with choosing one of these candidates as the output. There are many logically possible ways of doing this: pick the shortest candidate, pick the candidate derived with the fewest changes, etc. Optimality Theory makes this decision by balancing conflicting demands not to change the input and to have well-formed outputs. These demands are modeled using *constraints*, functions that assign penalties or *violations* to candidates. Like GEN, Prince and Smolensky (1993/2004:6) propose that the constraint set CON is “maximally universal”: all languages share a substantial set of constraints, allowing for idiosyncratic exceptions; see Ellison (2000), Hale and Reiss (2008), and Hayes and Wilson (2008) for critical discussion. This section discusses what constraints are and the following section covers how they interact.

#### 3.1 Faithfulness constraints

*Faithfulness constraints* penalize differences between the input and the candidate. Standardly, they are defined as counting the number of changes and assigning one violation for

each. In Correspondence Theory, there are three archetypal families of constraints that appear in most analyses. Two, the constraints MAX and DEP, are defined in (10) and (11), respectively. For each, I have provided a formal definition following McCarthy’s (2008a:175) format that specifies exactly how to assign violations and an informal definition. Analogous constraints can be defined over other correspondence relationships.

(10) MAX (McCarthy and Prince, 1994:342)

Assign one violation for every segment in the input without a correspondent in the candidate.

“All underlying segments are present in the candidate.”

(11) DEP (McCarthy and Prince, 1994:342)

Assign one violation for every segment in the candidate without a correspondent in the input.

“All segments in the candidate are underlying.”

Four candidates generated from /s’i-əto/ ‘even if one uses’ are presented in (12) along with the number of violations MAX assigns them. The *loci* of violation (McCarthy, 2002a, 2003a,b) are also shown to be as explicit as possible. In the fully faithful candidate (12a), all underlying segments are present, and MAX assigns zero violations; to reduce visual clutter, I’ve left this cell blank instead of writing a zero. Because it incurs zero violations, this candidate can be said to *satisfy* MAX. The remaining *unfaithful* candidates each incur at least one violation (12b-d): in candidates (12b-c), one underlying vowel is not present, and in candidate (12d), the three-segment suffix is not present. While I choose to represent violation totals with numbers, it is also common to represent them with strings of asterisks, e.g., \*\*\*. As the words *satisfy* and *violation* imply, fewer violations are better. Thus, among these four candidates, as far as MAX is concerned, candidate (12a) is better than candidates (12b) and (12c), which are not differentiated and are in turn better than candidate (12d). For MAX, the worst possible candidates are those in which none of the underlying segments are present, e.g., [.mæk.], which incur as many violations as there are underlying segments.

(12) Evaluation by MAX

| /s’i-əto/<br>1 2 3 4 5      | MAX                  |
|-----------------------------|----------------------|
| a. [s’i.ə.to.]<br>1 2 3 4 5 |                      |
| b. [s’ə.to.]<br>1 3 4 5     | 1 /i/<br>2           |
| c. [s’i.to.]<br>1 2 4 5     | 1 /ə/<br>3           |
| d. [s’i.]<br>1 2            | 3 /ə, t, o/<br>3 4 5 |

To illustrate DEP with concrete examples, consider the words in (13), which were borrowed from English into Korean, where they acquired more vowels. If these vowels are not underlying, they incur violations of DEP, as in (14). In the fully faithful candidate (14a), only underlying segments are present, and DEP is satisfied. Candidate (14b) incurs one violation for its non-underlying vowel, candidate (14c) incurs two, and candidate (14d) incurs



three. Unlike MAX, whose violations are bounded by the number of segments in the input, there is no maximum number of violations of DEP and, consequently, there is no worst candidate as far as DEP is concerned.

(13) English words borrowed into Korean (Kim-Renaud, 2022:43)

- a. [k<sup>h</sup><sub>1</sub>.r<sub>2</sub>i.s<sub>3</sub>i.ma.s<sub>4</sub>i.] ‘Christmas’
- b. [.a.i.s<sub>1</sub>i.k<sup>h</sup><sub>2</sub>i.rim.] ‘ice-cream’
- c. [.s<sub>1</sub>i.t<sup>h</sup><sub>2</sub>i.ra.i.k<sup>h</sup><sub>3</sub>i.] ‘strike’

(14) Evaluation by DEP

| /st <sup>h</sup> raik <sup>h</sup> /                                                                             | DEP                          |
|------------------------------------------------------------------------------------------------------------------|------------------------------|
| <sub>1 2 3 4 5 6</sub>                                                                                           |                              |
| a. [.st <sup>h</sup> <sub>1</sub> ra. <sub>2</sub> ik <sup>h</sup> <sub>3</sub> .]                               |                              |
| b. [.s <sup>h</sup> <sub>1</sub> i.t <sup>h</sup> <sub>2</sub> .ra. <sub>3</sub> ik <sup>h</sup> <sub>4</sub> .] | 1 [i] <sub>7</sub>           |
| c. [.s <sub>1</sub> i.t <sup>h</sup> <sub>2</sub> i.ra. <sub>3</sub> ik <sup>h</sup> <sub>4</sub> .]             | 2 [i, i] <sub>7 8</sub>      |
| d. [.s <sub>1</sub> i.t <sup>h</sup> <sub>2</sub> i.ra.i.k <sup>h</sup> <sub>3</sub> i.]                         | 3 [i, i, i] <sub>7 8 9</sub> |

One may object to describing candidate (14a) as fully faithful – after all, it has syllable structure while the input does not. Fully faithful candidates, however, are not those that are identical to the input, but rather those that do not violate any faithfulness constraints. It is standardly assumed that prosodification operations like syllabification and footing do not violate any faithfulness constraints (McCarthy, 2003c; McCarthy and Pruitt, 2013). In part, this is to prevent the grammar from modeling contrastive syllabification, e.g., [.sa.rə.] vs. [.sar.ə.], which is unattested; apparent counterexamples like Blackfoot (b1a; Algic; Elfner, 2006a,b) and Scottish Gaelic (g1a; Indo-European; Blevins, 1995) can be analyzed without contrastive syllabification (Elfner, 2006a,b; McCarthy, 2003c). Hence, candidates derived just by prosodifying the input are considered to be fully faithful.

The third common constraint archetype IDENT is defined in (15). IDENT constraints penalize featural differences (see Zoll, 1996 and McCarthy, 2008b for discussion) and are accordingly defined with reference to a specific feature; the definition below uses the generic F as a placeholder. Unlike MAX and DEP, IDENT constraints are often, but not always (see Pater, 1996, 1999, 2001 and Itô and Mester, 2003) defined symmetrically with respect to the input and output. Because they are defined over pairs of corresponding segments, IDENT constraints do not assign violations to segments absent from underlying representations or candidates.

(15) IDENT(F) (McCarthy and Prince, 1994:340)

Assign one violation for every pair of corresponding segments that differ in the feature F.

“All corresponding segments have the same value of the feature F.”

The laryngeal alternations in (16) provide concrete examples to illustrate evaluation by IDENT(lar), which penalizes corresponding segments with different laryngeal features (Lombardi, 1999, 2001). The fully faithful candidate in (17) by definition does incur any violations

of IDENT(lar) (17a). Candidate (17b) tenses the coronal stop, incurring one violation, candidate (17c) also aspirates the sibilant, incurring a second violation, and candidate (17d) voices the dorsal stop, incurring a third violation. In the unfaithful candidates, the corresponding sibilants also have different values for the feature [anterior], violating another IDENT constraint.

(16) Laryngeal alternations in Korean (Kang et al., 2022:218)

- a. /sik-tan/    [ʃik.t'an.]    ‘menu’
- b. /pak-t͡sa/    [pak.t͡s'a.]    ‘tempo’
- c. /k'ʌk'-t͡a/    [k'ʌk'.t'a]    ‘break off’

(17) Evaluation by IDENT(lar)

| /sik-tan/<br>123 456                    | IDENT(lar)                                       |
|-----------------------------------------|--------------------------------------------------|
| a. [sik.tan.]<br>123 456                |                                                  |
| b. [ʃik.t'an.]<br>123 456               | 1 /t/ → [t']                                     |
| c. [ʃ <sup>h</sup> ik.t'an.]<br>123 456 | 2 /s/ → [ʃ <sup>h</sup> ], /t/ → [t']            |
| d. [ʃ <sup>h</sup> ig.t'an.]<br>123 456 | 3 /s/ → [ʃ <sup>h</sup> ], /k/ → [g], /t/ → [t'] |

Like DEP, the number of violations of IDENT is unbounded. This is due to fission, an operation that splits an input segment into multiple segments, e.g., [sik.t͡a.n̩̥]. As in this example, the nasal in the input corresponds to multiple segments in the candidate. Under Freedom of Analysis, this can apply arbitrarily many times, incurring arbitrarily many violations of IDENT. Fission violates a family of faithfulness constraints called INTEGRITY (McCarthy and Prince, 1999; Yu, 2005; Saba Kirchner, 2010; Staroverov, 2014), one of many constraint families beyond the scope of this chapter; see McCarthy and Prince (1999) for a catalog of faithfulness constraints within Correspondence Theory.

In addition to the general faithfulness constraints defined above, which assign violations for every application of a given operation, OT analyses often utilize *positional faithfulness constraints* which only regulate specific environments. These include morphological roots (McCarthy and Prince, 1995), morphological stems (Baković, 2000, 2003, 2005b), syntactic heads and parts of speech (Moreton et al., 2017), the beginnings of words (Casali, 1997; Beckman, 1998), the ends of tiers (Jurgec, 2011; Bennett, 2015; Lamont, 2022a), stressed syllables (Beckman, 1998; Becker and Jurgec, 2020), syllable onsets (Beckman, 1998; Lombardi, 1999), and other prominent positions (Smith, 2005; see Dinnsen and Farris-Trimble, 2009 for a developmental perspective). How positional faithfulness constraints behave in OT and related frameworks has been a productive area of research (Jesney, 2011, 2016a,b; McCarvel and Kaplan, 2019).

## 3.2 Phonotactic constraints

*Phonotactic constraints* penalize illicit structures in candidates. Like faithfulness constraints, they are standardly defined as counting the number of offending structures and assigning

one violation for each. Phonotactic constraints are more often referred to as *markedness constraints*, following Prince and Smolensky (1993/2004:5). However, there are good reasons to avoid this terminology. For one thing, *markedness* is ambiguous and often refers to unrelated concepts (Haspelmath, 2006; Rice, 2007; Hume, 2011). For another, hierarchical relations between segments are also formally implemented by faithfulness constraints (de Lacy, 2002, 2006, 2011). For example, faithfulness constraints are responsible for the fact that Korean both inserts and deletes the high central vowel [ɨ] instead of another vowel. The vowel [ɨ] is unmarked relative to others in the sense that it is the least faithful.

In OT, unfaithful mappings are primarily motivated by phonotactic constraints (see Alderete, 1999 and Mortensen, 2006 for alternative motivations). In this sense, the framework models input-output mappings teleologically: /A/ surfaces as [B] because [B] is phonotactically more well-formed than [A]. This output orientation contrasts with the input orientation of many rule-based models, where, for example, /A/ surfaces as [B] because its underlying form met the structural description of a rewrite rule. One analytic advantage is that OT can formally unify processes that rule-based models cannot. A famous illustrative instance manifests in the morphophonology of Yowlumne Yokuts (yok; Yok-Utian; Kisseberth, 1969, 1970, 2011; see Kiparsky, 1973 for critical discussion). At least three processes have the effect of preventing triconsonantal clusters from surfacing, obeying a surface true generalization that complex onsets and clusters are illicit (Kuroda, 1967; Archangeli, 1984, 1991; Noske, 1985, 1992; Kisseberth, 2011; cf. Blevins, 2003); see Guekguezian (2011) and Adisasmito-Smith et al. (2023) for discussion of another dialect of Yokuts.

One instantiation of this generalization is that suffixes with initial consonant clusters surface without their first consonant after consonant-final stems (18); schematically /C-CCV/ → [C-CV]. The examples in (18a) illustrate this with the causative-repetitive suffix /-l̥sa:/ and those in (18b) with the passive consequent adjunctive /-h̥ne:l/. A second relevant process inserts vowels between root-final consonants when a consonant-initial suffix is added (19); schematically /CC-CV/ → [CVC-CV]. In (19a), the root /ʔilk<sup>h</sup>/ ‘to sing’ surfaces faithfully with the future suffix /-iɲ̥/ (19a.i) and with an additional vowel with the aorist /-hiɲ̥/ (19a.ii). In (19b), the root /p<sup>h</sup>aʔt<sup>h</sup>/ ‘to fight’ surfaces faithfully with the reciprocal-reflexive /-iwiɲ̥/ (19b.i) and with an additional vowel with the passive future /-ɲ̥it̥<sup>h</sup>/ (19b.ii) and the causative-repetitive (19b.iii). Lastly, repetition is exponed by reduplicating verbs with two consonants (20a) and with a suffix for verbs with three (20b). For other examples of phonotactic structures that motivate diverse processes, see Pater (1996, 1999, 2001), Cassimjee and Kisseberth (1999), Davis (2000), Lombardi (2001), Baković (2005a, 2016), Rubach (2007, 2011, 2014, 2023), Alderete and MacMillan (2015), Bennett (2015), Guekguezian (2016), and Zuraw (2018).

(18) Consonant deletion in Yowlumne (Newman, 1944:95, 166)

- a. [-lsa: ~ -sa:]
- i. /nine:ls̩a:hi<sub>n</sub> na? t<sup>h</sup>an/
  - [ni.ne<sub>l</sub>.sa:hi<sub>n</sub>. na?. t<sup>h</sup>an.]
  - ‘I got him to keep still several times’
  - ii. /ʔop<sup>h</sup>o:t<sup>h</sup>ls̩a:ni<sup>h</sup> na?/
  - [ʔo.p<sup>h</sup>et<sup>h</sup>.sa:ni<sup>h</sup>. na?.]
  - ‘I’ll be made to get up repeatedly’
- b. [-hnel ~ -nil]
- i. /wa<sub>n</sub>hi<sub>n</sub> ʔaman na<sub>n</sub> t’ik’i:hn̩e:l-ni ʔamin/
  - [wa<sub>n</sub>.hi<sub>n</sub>. ʔa.ma<sub>n</sub>. na<sub>n</sub>. t’i.k’eh.n̩e<sub>l</sub>.ni. ʔa.mi<sub>n</sub>.]
  - ‘they gave me something to tie him with’
  - ii. /ʔa:malhn̩e:l wi<sub>t</sub>e:pi<sub>n</sub>/
  - [ʔa.ma<sub>l</sub>.n̩i<sub>l</sub>. wi<sub>t</sub>e:pi<sub>n</sub>.]
  - ‘the place of the child’s being aided’

(19) Vowel epenthesis in Yowlumne (Newman, 1944:27, 33, 132, 150)

- a. [Ø ~ i]
- i. /ʔilk<sup>h</sup>i:n/
  - [ʔil.k<sup>h</sup>en.]
  - ‘(he) will sing’
  - ii. /ʔilk<sup>h</sup>hi<sub>n</sub>/
  - [ʔi.li<sup>h</sup>.hi<sub>n</sub>.]
  - ‘(he) sang’
- b. [Ø ~ i ~ e]
- i. /hi<sub>n</sub>a? maʔan ʔas p<sup>h</sup>aʔt<sup>h</sup>iwi<sub>s</sub>-a ʔohjoxoʔ/
  - [hi<sub>n</sub>.a?. .ma.ʔa<sub>n</sub>. ʔas. .p<sup>h</sup>aʔ.t<sup>h</sup>i:sa. ʔoh.jo.xoʔ.]
  - ‘perhaps you (plu.) indeed want to fight with one another’
  - ii. /miʔi<sub>n</sub> tiʔ maj p<sup>h</sup>aʔt<sup>h</sup>ni<sup>h</sup>/
  - [mi.ʔi<sub>n</sub>. .tiʔ. .maj. .p<sup>h</sup>a.ʔit<sup>h</sup>.ni<sup>h</sup>.]
  - ‘now, it is said, we will be fought’
  - iii. /p<sup>h</sup>aʔt<sup>h</sup>ls̩a:/
  - [p<sup>h</sup>a.ʔet<sup>h</sup>.sa:.]
  - ‘cause to fight repeatedly’

(20) Repetitive allomorphy in Yowlumne (Newman, 1944:66, 84, 89, 91, 109)

a. Reduplication

i. [ts<sub>1</sub><sub>2</sub><sub>3</sub>'omts<sub>4</sub><sub>5</sub><sub>6</sub>'u.mun<sub>7</sub>.hun<sub>8</sub>.]

‘(it) was destroyed repeatedly’

ii. [ts<sub>1</sub><sub>2</sub><sub>3</sub>'u.tu<sub>4</sub><sub>5</sub><sub>6</sub><sub>7</sub>ts<sub>8</sub><sub>9</sub><sub>10</sub>'tu<sub>11</sub>.sa.xo<sub>12</sub>?. ?a.mak<sub>13</sub>'.]

‘they (dual) are pulling each other’s hair repeatedly’

iii. [muh<sub>1</sub><sub>2</sub><sub>3</sub>muh<sub>4</sub><sub>5</sub><sub>6</sub>.la<sub>7</sub>t<sub>8</sub><sup>h</sup>.]

‘(he) was made to dive repeatedly’

b. /-ta:/

i. [.ʔi.p<sup>h</sup>at<sup>h</sup>.ta<sub>1</sub>.li<sub>2</sub>.]

‘that which has been lost many times’

ii. [.mi.ʔin<sub>1</sub>. .t<sup>h</sup>an<sub>2</sub>. .so.tox.to<sub>3</sub>?. .k<sup>h</sup>ew<sub>4</sub>. .si.liw<sub>5</sub>.]

‘now (he) will throw him repeatedly on these rocks’

iii. [.su.ta<sub>1</sub>k<sub>2</sub>'ta<sub>3</sub>.hin<sub>4</sub>. .su.t<sup>h</sup>o<sub>5</sub>.sol<sub>6</sub>. ?a.min<sub>7</sub>.]

‘(he) often removed his shirt’

The definition of a phonotactic constraint relevant to the syllabic analysis of Yowlumne Yokuts is provided in (21). Its evaluation of four candidates generated from /ʔilk<sup>h</sup>-hin/ ‘(he) sang’ is illustrated in (22). The two fully faithful candidates (22a-b) each incur one violation of \*COMPLEX; the first for its initial syllable with a coda cluster (22a) and the second for its second syllable with an onset cluster (22b). The unfaithful candidates (22c-d) satisfy \*COMPLEX because none of their syllables dominate complex constituents. In candidate (22c), a non-underlying vowel is present between the root consonants, and in candidate (22d), the suffix-initial consonant is absent. These candidates violate DEP and MAX, respectively, but I have excluded them for illustrative purposes; I discuss constraint interaction in the following section.

(21) \*COMPLEX (Prince and Smolensky, 1993/2004:96)

Assign one violation for every syllable in the candidate with more than one segment in the onset, nucleus, or coda.

“Syllable constituents only dominate one segment.”

(22) Evaluation by \*COMPLEX

| /ʔilk <sup>h</sup> -hin/<br><sub>1 2 3 4 5 6 7</sub>       | *COMPLEX                                       |
|------------------------------------------------------------|------------------------------------------------|
| a. [.ʔilk <sup>h</sup> .hin.]<br><sub>1 2 3 4 5 6 7</sub>  | 1 [.ʔilk <sup>h</sup> .]<br><sub>1 2 3 4</sub> |
| b. [.ʔil.k <sup>h</sup> hin.]<br><sub>1 2 3 4 5 6 7</sub>  | 1 [.k <sup>h</sup> hin.]<br><sub>4 5 6 7</sub> |
| c. [.ʔi.li <sup>h</sup> .hin.]<br><sub>1 2 3 4 5 6 7</sub> |                                                |
| d. [.ʔil.k <sup>h</sup> in.]<br><sub>1 2 3 4 6 7</sub>     |                                                |

The unfaithful candidates in (22) illustrate the shared phonotactic functions of consonant deletion and vowel epenthesis in Yowlumne Yokuts. If underlying CCC strings were to surface faithfully, they would necessarily incur a violation of \*COMPLEX (Yowlumne Yokuts does not allow syllabic nuclei, which is enforced by a different constraint). Unfaithful mappings that

prevent their surfacing, either by deleting a consonant or inserting a vowel, satisfy \*COMPLEX. Both are attested strategies deployed by the language in different morphophonological contexts. McCarthy (2002b:25) dubs this feature of OT *homogeneity of target/heterogeneity of process*: there are multiple logically possible ways to avoid an illicit structure, all of which are formally unified as satisfying a phonotactic constraint. Input-oriented frameworks like rewrite rules cannot formally express this unity (Kisseberth, 1969, 1970, 2011).

Allowing GEN to produce arbitrarily long candidates, there is no upper bound on the number of violations \*COMPLEX, or in principle any phonotactic constraint, can assign.

Like faithfulness constraints, there are families of phonotactic constraints. For example, the constraint \*COMPLEX in (21) is more often broken down by subsyllabic constituent: \*COMPLEXONSET, \*COMPLEXNUCLEUS, \*COMPLEXCODA (see Zec, 2007 for an overview).

One important family of phonotactic constraints is the set of ALIGN constraints, defined generically in (23) (McCarthy and Prince, 1993a; McCarthy, 2003b; Hyde, 2012, 2016). ALIGN constraints penalize non-contiguous structures or *subsequences* of the form Cat1...Cat3...Cat2 or Cat2...Cat3...Cat1. By minimizing the number of intervening Cat3s, they encourage the other two categories to be as close together as possible.

(23) ALIGN(Cat1, Edge1, Cat2, Edge2, Cat3) (McCarthy, 2003b:78)

For every Cat1, if there is a Cat2, assign one violation for every Cat3 that separates Edge1 of Cat1 from Edge2 of the nearest Cat2, where Cat1 and Cat2 are phonological or morphological objects, Cat3 is a prosodic object, and Edge1 and Edge2 are left or right edges.

“There are no Cat3s between Cat1s and Cat2s.”

As an illustration, consider stress in həŋqəmiŋəŋ, a dialect of Halkomelem (hur; Salish; Shaw et al., 1999; Suttles, 2004); the related dialect Hul’q’umin’um’ exhibits a similar pattern (Bianco, 1996, 1998) as do Northern dialects of Lushootseed (lut; Salish; Hess, 1977; Kye, 2023). The default pattern is for primary stress to fall on the initial syllable of the word, but full vowels (i.e., those other than [ə]) attract stress (Suttles, 2004:13). There are non-trivial interactions with the morphology (see Czaykowska-Higgins, 1993 for discussion of the complex interactions in a related language) and numerous lexical exceptions to this generalization, such as [ʔi.(ʔə.qə).tsəs.] \*[(ʔi.ʔə).qə.tsəs.] ‘five canoes’ (Suttles, 2004:465). As in this example, I follow Shaw (2009) in analyzing həŋqəmiŋəŋ as having trochaic feet. The default stress pattern is illustrated by the words in (24) and (25); I want to emphasize that although these examples provide a thorough data set, their morphological structure plays a non-trivial role. In (24), words without full vowels have initial stress by default. When there is only one full vowel in the word, it is stressed regardless of its position in the word (25).

(24) həŋqəmiŋəŋ words without full vowels (Suttles, 2004:24, 38, 48, 50)

- a. [(ʔ.səq’.)] ‘split, tear’
- b. [(ʔ.ʔə.q’əʔ.)] ‘usually-PST’
- c. [(ʔ.kʷə.nə).təs.] ‘get.taken-TR-3TR’
- d. [(ʔ.ʔəʔ.tə).nəs.təxʷ.] ‘eat-CAUS’

(25) həŋqəminəm words with one full vowel (Suttles, 2004:24, 28, 37, 39, 49-50, 94, 111)

- a. i. [(.'p'etθ'.)] 'sew'
- b. i. [(.'tse.ləx.)] 'hand'
- ii. [.x<sup>w</sup>ə.(.'θi.)] 'become-big'
- c. i. [(.'x<sup>w</sup>q<sup>w</sup>e.nəts.).təs.] '[she] punches holes in the bottom of it'
- ii. [.tsəl.(.'tse.ləx.)] 'hands'
- iii. [.χ<sup>w</sup>ə.χ<sup>w</sup>ə.(.'ʔjem.)] 'be.story.telling'
- d. i. [(.'k<sup>w</sup>'e.k<sup>w</sup>'ə.).jə.təs.] 'be.training-TR-3TR'
- ii. [.nə.(.'ʔe.məs.).təx<sup>w</sup>.] 'go-CAUS'
- iii. [.wə.x<sup>w</sup>ə.(.'ne:n.təs.)] 'when-become-becoming.night-3SUB'
- iv. [.x<sup>w</sup>əs.θəl.θə.(.'leq.)] 'EST-become-separated(PL.RES)'

There are two conflicting pressures in həŋqəminəm: one is for feet to be as far to the left as possible, and the other is for stress to be on a full vowel. To illustrate ALIGN constraints, I focus on the former here, and return the latter in the following section. The constraint ALLFEETLEFT, defined in (26), penalizes feet not at the left edge of the word. Its evaluation of four candidates is illustrated in (27); though not explicitly marked, all four are taken to be parsed into prosodic words. Salish schwas are most commonly analyzed as epenthetic (Kinkade, 1998; Blake, 2000; Parker, 2011); I include them in the following underlying forms as a simplification, but that decision does not affect the analysis. In candidate (27a), the only foot's left edge is co-extensive with the left edge of the word, satisfying ALLFEETLEFT. In (27b), the foot is separated from the left edge of the word by one syllable, incurring one violation, and in (27c), the foot incurs two violations; to simplify presentation, I only include the intervening syllables instead of the entire violating subsequence. In (27d), every syllable except the first is parsed into its own foot, incurring a total of six violations. The figure in (28) explicitly illustrates the locus swarm in candidate (27d). Every foot incurs as many violations as there are syllables before it, resulting in a large number of overlapping subsequences that each violate ALLFEETLEFT.

(26) ALLFEETLEFT (McCarthy and Prince, 1993a:94)

For every foot, assign one violation for every syllable that separates the left edge of the foot from the left edge of the nearest word.

“There are no syllables separating feet from the left edge of the word.”

(27) Evaluation by ALLFEETLEFT

| /ʔəltən-stəx <sup>w</sup> /                    | ALLFEETLEFT                                             |
|------------------------------------------------|---------------------------------------------------------|
| <sub>1 2 3 4 5 6 7 8 9 10</sub>                |                                                         |
| a. [(.'ʔəlt.tə.).nəs.təx <sup>w</sup> .]       |                                                         |
| <sub>1 2 3 4 5 6 11 7 8 9 10</sub>             |                                                         |
| b. [.ʔəlt.(.'tə.nəs.).təx <sup>w</sup> .]      | 1 [.ʔəlt.]                                              |
| <sub>1 2 3 4 5 6 11 7 8 9 10</sub>             |                                                         |
| c. [.ʔəlt.tə.(.'nəs.təx <sup>w</sup> .)]       | 2 [.ʔəlt.], [.tə.]                                      |
| <sub>1 2 3 4 5 6 11 7 8 9 10</sub>             |                                                         |
| d. [.ʔəlt.(.'tə.)(.nəs.)(.təx <sup>w</sup> .)] | 6 [.ʔəlt.], [.ʔəlt.], [.ʔəlt.], [.tə.], [.tə.], [.nəs.] |
| <sub>1 2 3 4 5 6 11 7 8 9 10</sub>             | <sub>1 2 3 1 2 3 1 2 3 4 5 4 5 6 11 7</sub>             |

(28) Loci of violation of ALLFEETLEFT



### 3.3 Constraint preferences

The preceding discussion focuses on the mechanics behind constraints. Before turning to how they interact in the next section, it is important to discuss what role they play in the grammar. Constraints assign violations to candidates, and as the word implies, having more violations is worse. Put another way, a constraint *prefers* a candidate that has fewer violations than another. Of the candidates in (27), ALLFEETLEFT prefers candidate (27a) to candidates (27b-d), candidate (27b) to candidates (27c-d), and candidate (27c) to candidate (27d). This exactly mirrors the number of violations each candidate incurs:  $0 < 1 < 2 < 6$ . In OT, it is this preference relation that matters for the grammar, not necessarily the severity of violation. Consequently, constraints do not necessarily need to count violations, and many constraint types have been proposed that establish preferences in other ways.

One such proposal establishes preferences in terms of where loci appear in the candidate (Eisner, 2000a, 2002; Finley, 2008; Lamont, 2022a). *Directional constraints*, like counting constraints, prefer satisfying candidates to violating candidates. However, when comparing violating candidates, the location of the violation, not the total number, is what matters: a given directional constraint either prefers violations to be as far to the left as possible or as far to the right as possible. As an illustration, consider the phonotactic constraint \*STRESSED SCHWA, defined in (29) and its directional evaluation of the same candidates as (27) in (30); the indices indicate each segment’s position in the candidate rather than its correspondence to the input. The superscript arrow  $\Rightarrow$  means that this constraint prefers violations be as far to the right as possible – it evaluates *left-to-right*. \*STRESSED SCHWA $\Rightarrow$  prefers candidate (30c) to candidates (30a-b,d) because the latter all have violations closer to the left edge. It prefers candidate (30b) to candidates (30a,d) and prefers candidate (30d) to candidate (30a). A counting constraint would instead prefer (30a) to (30d) because the former has fewer total violations.

(29) \*STRESSED SCHWA (Cohn and McCarthy, 1998:76)

Assign one violation for every schwa that bears stress.

“There are no stressed schwas.”

(30) Evaluation by \*STRESSED SCHWA $\Rightarrow$

| /ʔəʔtən-stəxʷ/                                            | *STRESSED SCHWA $\Rightarrow$    |
|-----------------------------------------------------------|----------------------------------|
| a. [(ʔəʔ.tə.)(.nəs.təxʷ.)]<br>1 2 3 4 5 6 7 8 9 10 11     | 1<br>1 2 3 4 5 6 7 8 9 10 11     |
| b. [ʔəʔ.('tə.nəs.)(.təxʷ.)]<br>1 2 3 4 5 6 7 8 9 10 11    | 1<br>1 2 3 4 5 6 7 8 9 10 11     |
| c. [ʔəʔ.tə.('nəs.təxʷ.)]<br>1 2 3 4 5 6 7 8 9 10 11       | 1<br>1 2 3 4 5 6 7 8 9 10 11     |
| d. [ʔəʔ.('tə.)(.nəs.)(.təxʷ.)]<br>1 2 3 4 5 6 7 8 9 10 11 | 1 1 1<br>1 2 3 4 5 6 7 8 9 10 11 |



While directional constraints are superficially very different from counting violations, the underlying principle is the same. Counting constraints prefer candidates with a lower number of violations. Directional constraints do the same thing, but *lower* is defined in terms of *lexicographic order*. Because this concept is also relevant to section 4, it is worth spelling out in detail. In (31), I have organized the names of five language isolates taken from Campbell (2018) in lexicographic or *alphabetical* order: *Karuk* < *Kimki* < *Kwaza* < *Warao* < *Washo*. This is determined by comparing the names letter-by-letter from left to right. One name is lexicographically smaller than another if it is smaller in the leftmost position where they differ. Because *k* comes before *w* in the alphabet, we know that *Karuk*, *Kimki*, and *Kwaza* come before *Warao* and *Washo*. It does not matter what happens later in the names; only the leftmost position where they differ matters. The second letter tells us that *Karuk* comes before *Kimki* because *a* < *i* and that *Kimki* comes before *Kwaza* because *i* < *w*. The third letter tells us that *Warao* comes before *Washo* because *r* < *s*. As discussed in section 4, OT is standardly only concerned with identifying the lexicographically smallest object (*Karuk* < {*Kimki*, *Kwaza*, *Warao*, *Washo*}), but Coetzee (2004) argues that a full order (*Karuk* < *Kimki* < *Kwaza* < *Warao* < *Washo*) is useful for modeling variation. Turning back to the example in (30), if I replace the violations with the strings *abaaaaaaaa*, *aaaabaaaaaa*, *aaaaaabaaaa*, and *aaaababaaba*, where *a* and *b* stand for 0 and 1, respectively, and arrange them alphabetically, I derive exactly the same order as the preference relation: *aaaaabaaaa* < *aaaabaaaaaa* < *aaaababaaba* < *abaaaaaaaaa*.

(31) Five language isolates' names in alphabetical order

|   |   |   |   |   |
|---|---|---|---|---|
| K | a | r | u | k |
| K | i | m | k | i |
| K | w | a | z | a |
| W | a | r | a | o |
| W | a | s | h | o |

Other constraint types are defined directly in terms of their preferences, eschewing reference to ordered sets like numbers. *Targeted constraints* are a prime example of this (Wilson, 2000, 2001, 2003, 2013; Baković and Wilson, 2000). Informally, targeted constraints prefer that specified phonotactic violations are removed by applying a specified operation. For example, the targeted constraint DELETECOMPLEX defined in (32) penalizes complex syllable constituents like \*COMPLEX (21), but only prefers candidates that avoid them by deleting consonants. Of the candidates in (22), DELETECOMPLEX only prefers [ʔil.k<sup>h</sup>in.] (22d) to [ʔilk<sup>h</sup>.hin.] (22a). Candidate (22c) [ʔi.lik<sup>h</sup>.hin.] also satisfies \*COMPLEX, but because it does so by inserting a vowel, DELETECOMPLEX does not prefer it to the faithful candidates.

(32) DELETECOMPLEX (after Wilson, 2001:160)

Let *x* be any candidate and *α* be any consonant in *x* in a complex onset, nucleus, or coda. If candidate *y* is exactly like *x* except that *α* has been removed, then prefer *y* to *x*.

“Delete consonants to avoid complex syllables.”

### 3.4 Restrictions on CON

Standard OT constraints penalize either input-output mappings (faithfulness constraints) or outputs (phonotactic constraints). There is a third logical possibility, namely constraints on inputs, such as morpheme structure constraints (Halle, 1959; Booij, 2011). Prince and Smolensky (1993/2004:225) reject all restrictions on the input, adopting the hypothesis that any input is possible in any language, dubbed *Richness of the Base*. Thus, not only must the grammar of Korean generate the candidate [. $\text{O}\ddot{\text{u}}\text{g}$ .], it must also be equipped to deal with / $\text{O}\ddot{\text{u}}\text{g}$ / as an underlying representation. This hypothesis has been standardly assumed in the OT literature and productively incorporated in a number of analyses (Kurusu, 2000; McCarthy, 2005; Tessier and Jesney, 2014), but has been questioned on empirical and theoretical grounds (Hall, 2007; Tebay, 2022; Rasin, 2023; Gouskova, to appear).

Other restrictions on CON have largely focused on the syntax of constraints (Eisner, 1997b; Potts and Pullum, 2002; McCarthy, 2003b; de Lacy, 2011; Heinz, 2018). The preceding discussion presents constraints as functions associated with a set of penalized structures or relations. They scan through candidates assigning one violation to each structure or relation (see Lamont, 2022a for a formalization along these lines). One elaboration on this structure, *conjoined constraints*, defines constraints that assign violations only if a structure or relation simultaneously violates two specified constraints (Smolensky, 1993, 2006; Hewitt and Crowhurst, 1996; Moreton and Smolensky, 2002; Lubowicz, 2003a,b, 2005; Itô and Mester, 2003; Pater, 2009a; Crowhurst, 2011). For example, the conjoined constraint defined in (33) only penalizes syllables that violate both \*STRESSED SCHWA and \*COMPLEX: [. $\text{?}\text{ə}\text{lk}^{\text{h}}$ .] incurs a violation but [. $\text{?}\text{ilk}^{\text{h}}$ .] and [. $\text{?}\text{ə}\text{ł}$ .] do not. While conjoined constraints have been used widely, it has been argued to be unnecessary to combine constraints using other logical operators and shown that they make implausible predictions (Wolf, 2007; O’Hara, 2016; cf. Struijke, 2000).

(33) [\*STRESSED SCHWA & \*COMPLEX]/ $\sigma$

Assign one violation for every syllable in the candidate if (1) it has more than one segment in the onset, nucleus, or coda and (2) it is stressed and contains a schwa.

“No stressed syllables contain a complex constituent and schwa.”

Like GEN, computational approaches have restricted constraints to finite-state functions. This excludes logically possible but empirically implausible constraints from CON like PALINDROMIC defined in (34). This constraint is satisfied only by candidates that are their own mirror image, such as the  $\text{h}\text{ə}\text{n}\text{q}\text{ə}\text{m}\text{i}\text{n}\text{h}\text{ə}\text{m}$  words [(. $\text{'s}\text{q}\text{e}\text{q}\text{s}$ .)] ‘younger.sibling-3POS’ (Suttles, 2004:133) and [. $\text{w}\text{ə}$ .(. $\text{'n}\text{a}$ .. $\text{n}\text{ə}\text{w}$ .)] ‘EST-very’ (Suttles, 2004:456), ignoring prosodic structure. As discussed in section 2, finite-state functions cannot reverse strings, and it follows that identifying palindromes is also impossible.

(34) PALINDROMIC (Eisner, 1997b:1)

Assign one violation for every candidate that is not a palindrome.

“Candidates must be palindromes.”

Restricting constraints to finite-state functions also bans empirically motivated constraints such as the ALIGN family (Eisner, 1997b, 2000a, 2002). As mentioned earlier, constraints like ALLFEETLEFT evaluate overlapping structures and the number of violations it can assign to a candidate containing  $n$  syllables is on the order of  $n^2$ . This is impossible for a finite-state constraint, whose violations grow linearly in the length of a candidate (Biró, 2003). While this motivates removing ALIGN constraints, it does little to nip the computational problem in the bud. Counting constraints derive very complex patterns whether their violations grow linearly or quadratically (Lamont, 2021, 2022b). Finite-state OT grammars must eschew counting constraints entirely (Karttunen, 1998; Eisner, 2002) or impose a maximum number of violations (Frank and Satta, 1998).

## 4 Compromise: EVAL

It is the task of the evaluator function EVAL to take the set of candidates from GEN and the constraint set in CON and choose an output form (Prince and Smolensky, 1993/2004:5 originally called this function *H-eval*). This involves balancing the preferences of the constraints which often *conflict* with each other. Perhaps the most salient conflicts occur between phonotactic constraints which prefer well-formed outputs and faithfulness constraints which prefer unmodified inputs; without faithfulness constraints, OT grammars should only produce phonotactically perfect outputs (Chomsky, 2015:350). Some variants of Optimality Theory such as *Harmonic Grammar* (Legendre et al., 1990a,b; Pater, 2009b, 2016) and *MaxEnt* (Goldwater and Johnson, 2003; Jäger, 2007; Hayes, 2022; Wang and Wilson, [this volume](#)) adopt a democratic process whereby all constraints contribute to the decision, their relative influence being determined by their weight. OT instead imposes a strict hierarchy or *ranking* on the constraints and respects their preferences accordingly. As the previous sections discuss, GEN is taken to be universal and CON is taken to be (nearly) universal. The only substantial difference between languages, then, is their constraint ranking.

Recall that in həŋqəmiŋəŋ, stress is on the leftmost syllable unless it contains a schwa. This is modeled with two conflicting constraints on stress: feet should be as far to the left as possible, formalized as ALLFEETLEFT, and schwas should not be stressed, formalized as \*STRESSED SCHWA. From words like [.x<sup>w</sup>əs.θəl.θə.('leq.)] 'EST-become-separated(PL.RES)' (Suttles, 2004:111), we can infer that the latter's preferences are respected when they conflict with the former's. In an OT analysis of həŋqəmiŋəŋ stress, this is modeled by ranking \*STRESSED SCHWA above ALLFEETLEFT, written \*STRESSED SCHWA  $\gg$  ALLFEETLEFT.

This ranking is illustrated in the *tableau* in (35). OT tableaux represent input-output mappings; in this example, the underlying form /wə-x<sup>w</sup>ə-s-CəC-θəleq/ is given in the upper left cell and the surface form or *winner* [.x<sup>w</sup>əs.θəl.θə.('leq.)] (35a) is indicated with a pointing hand . Below it is a *losing* candidate [(.'x<sup>w</sup>əs.θəl.θə.leq.] (35b) with initial stress. The remaining columns correspond to the constraints \*STRESSED SCHWA and ALLFEETLEFT, with their left-to-right order reflecting their relative ranking; to save horizontal space, I have rotated the constraint names by ninety degrees. Cells at the intersection of a candidate's row and a constraint's column record that constraint's evaluation of that candidate and its preference (if any) relative to the winner. In this case, candidate (35a) satisfies \*STRESSED SCHWA and incurs three violations of ALLFEETLEFT, and candidate (35b) in-

curs one violation of \*STRESSED SCHWA and satisfies ALLFEETLEFT. The W means that \*STRESSED SCHWA prefers the winner to this loser and the L means that ALLFEETLEFT prefers this loser to the winner (Prince, 2002; Brasoveanu and Prince, 2011; Yanovich, 2014; Merchant and Prince, 2023). No cells in the winner’s row are labeled W or L because the winner cannot be preferred to itself.

(35) \*STRESSED SCHWA  $\gg$  ALLFEETLEFT (həŋqəmiŋəŋ)

|                                      |                                          |   |   |   |   |   |   |    |    |    |    |    |                 |   |               |  |
|--------------------------------------|------------------------------------------|---|---|---|---|---|---|----|----|----|----|----|-----------------|---|---------------|--|
| /wə-x <sup>w</sup> ə-s-CəC-θə l e q/ |                                          |   |   |   |   |   |   |    |    |    |    |    | *STRESSED SCHWA |   | ALL FEET LEFT |  |
| 1                                    | 2                                        | 3 | 4 | 5 | 6 | 7 | 8 | 9  | 10 | 11 | 12 | 13 |                 |   |               |  |
| ☞                                    | a. [.x <sup>w</sup> əs.θəl.θə.(.l e q.)] |   |   |   |   |   |   |    |    |    |    |    |                 | 3 |               |  |
|                                      | 3                                        | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 |    |                 |   |               |  |
|                                      | b. [(.x <sup>w</sup> əs.θəl.).θə.l e q.] |   |   |   |   |   |   |    |    |    |    |    | W 1             | L |               |  |
|                                      | 3                                        | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 |    |                 |   |               |  |

As discussed in the previous section, what matters is the constraints’ preferences, not the absolute number of violations. Under the ranking \*STRESSED SCHWA  $\gg$  ALLFEETLEFT, one violation of \*STRESSED SCHWA is strictly worse than arbitrarily many violations of ALLFEETLEFT. In order for a candidate to win, its row must be lexicographically smaller than all losing candidates’ rows. As before, this analogizes to alphabetical order: taking 0 violations to be *a*, 1 to be *b*, and so on, we can replace the two rows in the tableau in (35) with *ad* and *ba*, and observe that *ad* is indeed alphabetically smaller than *ba*. This is true regardless of *ad*’s second letter. For example, the word [nəx<sup>w</sup>.məθ.k<sup>w</sup>ə.jə.(.mi<sup>?</sup>ws.)] ‘a person originally from həŋqəmiŋəŋ but now living somewhere else’ (Suttles, 2004:282) incurs four violations of ALLFEETLEFT to avoid one violation of \*STRESSED SCHWA. In its alphabetical analogue, *ae* (0,4) is smaller than *ba* (1,0), as would be *af*, *ag*, and so on. When alphabetizing, the only comparison that matters is the leftmost position where two words differ. Exactly the same is true in OT: when comparing the winner to a loser, it must be case that the highest ranked constraint that distinguishes them prefers the winner. W/L notation provides a useful visual clue: in every loser’s row that contains an L, there must be a W to its left.

## 4.1 Typology

Given that OT only permits languages to differ in their constraint rankings, every analysis makes typological predictions. In həŋqəmiŋəŋ, \*STRESSED SCHWA is ranked above ALLFEETLEFT. By reversing the ranking, we predict a language with initial stress even when a non-schwa is available. This is the case in Comox-Sliammon (coo; Salish; Blake, 2000), which is closely related to Halkomelem. As in the words in (36), stress is on the initial syllable by default, even when a full vowel is available [(.tɬ’əp.x<sup>w</sup>a.).tʊɬtɬ.] \*[tɬ’əp.(.x<sup>w</sup>a.tʊɬtɬ.)] ‘I broke it’ (Blake, 2000:102).

(36) Stress in Comox-Sliammon (Blake, 2000:xxi, 22, 86, 160, 168, 172, 187, 211, 326, 332, 363)

- a. i. [(.'tθ'ok<sup>w</sup>.)] 'day, light, bright'  
 ii. [(.'p'əθk.)] 'bullhead (fish)'  
 b. i. [(.'k<sup>w</sup>'o.t'a.) ~ (. 'k<sup>w</sup>'u.t'a.)] 'barbecuing stick'  
 ii. [(.'t'ən.t'ɛn.)] 'barbecued fish'  
 iii. [(.'q<sup>w</sup>a.səm.)] 'flower'  
 iv. [(.'sə.χəm.) ~ (. 'sΛ.χΛm.)] 'racing canoe'  
 c. i. [(.'χaj.χe.).k<sup>w</sup>'us.] 'nightmare'  
 ii. [(.'mə.na.).tθ'ɛ.] 'drum'  
 iii. [(.'q<sup>w</sup>oq<sup>w</sup>.pə.).fi<sup>2</sup>n.] 'bit of hair on legs'  
 iv. [(.'so.so.).pə<sup>2</sup>m.] 'chopping (s.t.)'  
 v. [(.'θəm.θo).mən.] 'eyebrows'  
 vi. [(.'tɬ'əp.tɬ'ə.).pa.tɬ.] 'cedar bark baskets'  
 vii. [(.'dʒə'tɬ'.dʒə'tɬ').tθəm.] 'I'll go running'

The tableau in (37) illustrates the ranking in Comox-Sliammon. The winner (37a) satisfies ALLFEETLEFT because its foot is co-extensive with the left edge of the word (again, not marked explicitly but assumed) and incurs one violation of \*STRESSED SCHWA. The losing candidate that stresses the low vowel (37c) is what we would expect to win in hənqəminəm – it violates ALLFEETLEFT and satisfies \*STRESSED SCHWA – but under this ranking it loses. The other losing candidate has a stressed schwa in the middle of the word (37b), violating both constraints. There is no ranking under which this candidate wins because it has a strict superset of the violations of candidate (37a); it is a *perpetual loser* (Samek-Lodovici and Prince, 1999:1). In the alphabetical analogy, *ab* is smaller than *bb* whether you reverse the letters or not.

(37) ALLFEETLEFT ≫ \*STRESSED SCHWA (Comox-Sliammon)

|                                                                                                                                         | ALLFEETLEFT | *STRESSED SCHWA |
|-----------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------|
| /CəC-tɬ'apatiɬ/<br>1 2 3 4 5 6 7 8 9 10                                                                                                 |             |                 |
|  a. [(.'tɬ'əp.tɬ'ə.).pa.tɬ.]<br>1 2 3 4 5 6 7 8 9 10 |             | 1               |
| b. [tɬ'əp.(.'tɬ'ə.pa.).tɬ.]<br>1 2 3 4 5 6 7 8 9 10                                                                                     | W 1         | 1               |
| c. [tɬ'əp.tɬ'ə.(.'pa.tɬ.)]<br>1 2 3 4 5 6 7 8 9 10                                                                                      | W 2         | L               |

With  $n$  constraints, there are  $n!$  total possible rankings. The tableaux in (35) and (37) illustrate both of the  $2! = 2$  rankings of ALLFEETLEFT and \*STRESSED SCHWA. As the constraint set grows, the typological space explodes. For example, in one of few OT analyses of an entire language, Cahill (2007) uses 127 constraints to model the phonology of Konni (kma; Niger-Congo), implying a factorial typology comprising  $3.0 \times 10^{213}$  total rankings. Not every ranking produces a distinct grammar, however (see Anttila, 1997, 2002 and Anttila and Cho, 1998 for discussion of the role this plays in variation), and typological work in OT aims to distill the set of predicted grammars from a constraint set.

Given their rate of growth, OT typologies are dubbed *factorial typologies* (Prince and Smolensky, 1993/2004:33; Gordon, 2007). For examples, see Prince and Smolensky (1993/2004), Morén (1999), and Zec (2007) for factorial typologies of syllable structure and Gordon (2002), Buckley (2009), Kager (2012), Pruitt (2012), Martínez-Paricio and Kager (2015), and Lamont (2023) for factorial typologies of footing. Factorial typologies are powerful ways of predicting what are possible and impossible languages and a crucial factor that distinguishes OT from other models like Chomsky and Halle’s (1968) rule-based formalism. The hypothesis that CON is maximally universal implies from the analysis of any given language predictions about every possible language. In models with entirely language-specific components, any cross-linguistic similarity is accidental, and there is no ability to make fine-grained typological predictions (see Prince, 2007 for discussion along these lines).

## 4.2 Learning

Given the vast grammatical spaces imposed by factorial growth, a reasonable question to ask is how a child learns the constraint ranking of their ambient language. Constraint rankings are known to be learnable from the appropriate winner~loser comparisons (Tesar and Smolensky, 2000; Brasoveanu and Prince, 2011), and there is an empirically testable theory of acquisition couched in OT. In the initial state of the grammar, all phonotactic constraints are ranked above all faithfulness constraints (Demuth, 1995; Smolensky, 1996; Pater, 1997; Gnanadesikan, 2004), resulting in outputs that are maximally phonotactically well-formed, but possibly unfaithful; cf. Hale and Reiss (1998) who argue for the opposite initial ranking. Over time, children adjust their rankings to allow surface forms to violate more phonotactic constraints faithfully, passing through intermediate grammars until they achieve an adult-like grammar. For example, Dutch (nld; Indo-European) syllable structure (C)(C)V(C)(C) allows onsetless syllables [.a.pi.] ‘monkey’ (Levelt et al., 2000:246), closed syllables [.pus.] ‘cat’ (Levelt et al., 2000:247), and syllables with complex margins [.xrɔnt.] ‘floor’ (Levelt et al., 2000:257). While these three structures violate phonotactic constraints, .CV. syllables do not. Levelt et al. (2000) demonstrate that along the acquisitional paths children follow when learning Dutch, they start by producing .CV. syllables before acquiring codas, empty onsets, and finally complex margins, consistent with the gradual reranking hypothesis.

## 4.3 An illustrative analysis

In general, given a phonotactic constraint  $\mathbb{P}$  and a faithfulness constraint  $\mathbb{F}$ , the grammar models either neutralization  $\mathbb{P} \gg \mathbb{F}$  or contrast  $\mathbb{F} \gg \mathbb{P}$ . Both mappings may be blocked by a

higher ranked constraint  $\mathbb{B}$  (short for Blocker). All four ranking possibilities are illustrated by labialization allophony in Nuu-chah-nulth (nuk; Wakashan); see Sapir and Swadesh (1939), Rose (1981), Stonham (1999), Kim (2003), and Werle (2010) for discussion and Stonham and Kim (2008) for a phonetic study.

In Nuu-chah-nulth, dorsal stops contrast in the feature [round]: compare [...ak'a:...] (38a.i) and [...ak<sup>w</sup>'a...] (38b.i). This contrast is neutralized word-finally to [−round] [...ak#] (38a.ii, b.ii) and after round vowels to [+round] [...uq<sup>w</sup>i:...] (38a.iii) [...u:k<sup>w</sup>a...] (38b.iii). Word-finally after a round vowel, only [−round] stops surface, as in [...uk#] (38a.iv). Stonham (1999) notes that final dorsal stops after [u] are phonetically labialized, as in [he:ʔjɪʔs.ʔat.tuk<sup>wh</sup>.] ‘they were drinking his blood’ (Stonham, 1999:141), but analyzes these as phonologically [−round].

(38) Labialization allophony in Tsishaath Nuu-chah-nulth (Stonham, 1999:22, 70, 76, 86, 95, 133)

a. Underlyingly [−round] dorsal stops

- |                                  |                                           |                                   |
|----------------------------------|-------------------------------------------|-----------------------------------|
| i. /ʔu-ʔju:ʔa-ʔak-ʔa:h-ʔatʔ-ʔim/ | [.ʔu.ʔju:k.ʔa.k'a:h.ʔa.tʔ'im.]            | ‘name it after that’              |
| ii. /ʔah-ʔin-ʔak/                | [.ʔah.ʔin.ʔak.]                           | ‘his diver’                       |
| iii. /tʃʔitʃʔmuq-i:ʔa:qtʔ-qa/    | [.tʃʔitʃ.ʔmu.q <sup>w</sup> i:ʔa:qtʔ.qa.] | ‘they will make him a scape-goat’ |
| iv. /meʔitʔqats-ʔis-uk/          | [.mɛ.ʔitʔ.qats:.ʔis.suk <sup>h</sup> .]   | ‘boy-DIM-POSS’                    |

b. Underlyingly [+round] dorsal stops

- |                                               |                                            |                             |
|-----------------------------------------------|--------------------------------------------|-----------------------------|
| i. /qaḥ-ak <sup>w</sup> -ʔas/                 | [.qa.ḥa.k <sup>w</sup> as.]                | ‘on the ground unconscious’ |
| ii. /ʔiʔa:-ak <sup>w</sup> /                  | [.ʔi.ʔa:ʔak.]                              | ‘missed him’                |
| iii. /CV-tʃuk <sup>w</sup> a:-ʔin-mas-mit-ʔi/ | [.tʃu.tʃu:k <sup>w</sup> a.ʔin.ma.sit.ʔi.] | ‘the inviters’              |

Four constraints are needed to model this allophony: IDENT(round), which penalizes changes to the feature [round], \*C<sup>w</sup>, which penalizes labialized stops, \*uC[−round], which penalizes non-labialized consonants after [u], and \*C<sup>w</sup>#, which penalizes labialized consonants word-finally.

Ranking IDENT(round) above \*C<sup>w</sup> ( $\mathbb{F} \gg \mathbb{P}$ ) derives the surface contrast between [+round] and [−round] stops. The tableau in (39) illustrates this ranking: both stops surface faithfully (39a) as changing either violates IDENT(round) (39b-c). To simplify the presentation, I only consider candidates derived by changing the feature [round].

(39) IDENT(round)  $\gg$  \*C<sup>w</sup> (Nuu-chah-nulth)

|                                                                |              |                 |
|----------------------------------------------------------------|--------------|-----------------|
|                                                                | IDENT(round) |                 |
| /qaḥ-ak <sup>w</sup> -ʔas/<br>1 2 3 4 5 6 7 8                  |              | *C <sup>w</sup> |
| ☞ a. [.qa.ḥa.k <sup>w</sup> .as.]<br>1 2 3 4 5 7 8             |              | 1               |
| b. [.qa.ḥa.k <sup>ʰ</sup> .as.]<br>1 2 3 4 5 7 8               | W 1          | L               |
| c. [.q <sup>w</sup> a.ḥa.k <sup>w</sup> .as.]<br>1 2 3 4 5 7 8 | W 1          | W 2             |

Ranking \*uC[−round] above IDENT(round) neutralizes the contrast between [+round] and [−round] stops after [u] ( $\mathbb{P} \gg \mathbb{F}$ ), blocking the faithful realization of [−round] stops ( $\mathbb{B} \gg \mathbb{F} \gg \mathbb{P}$ ). The tableau in (40) illustrates this effect: the /q/ in the noun is labialized (40b), satisfying \*uC[−round] at the expense of the two lower ranked constraints.

(40) \*uC[−round]  $\gg$  IDENT(round) (Nuu-chah-nulth)

|                                                                                                                |             |              |                 |
|----------------------------------------------------------------------------------------------------------------|-------------|--------------|-----------------|
|                                                                                                                | *uC[−round] | IDENT(round) | *C <sup>w</sup> |
| /tʃ <sup>h</sup> itʃ <sup>h</sup> muq-iḥ-ʔa:qtḥ-qa/<br>1 2 3 4 5 6 7 8 9 10 11 12 13 14                        |             |              |                 |
| a. [.tʃ <sup>h</sup> itʃ <sup>h</sup> .ʔmu.qiḥ.ʔa:qtḥ.qa.]<br>1 2 3 4 5 6 7 8 9 10 11 12 13 14                 | W 1         | L            | L               |
| ☞ b. [.tʃ <sup>h</sup> itʃ <sup>h</sup> .ʔmu.q <sup>w</sup> iḥ.ʔa:qtḥ.qa.]<br>1 2 3 4 5 6 7 8 9 10 11 12 13 14 |             | 1            | 1               |

Finally, \*C<sup>w</sup># blocks \*uC[−round] from triggering labialization word-finally ( $\mathbb{B} \gg \mathbb{P} \gg \mathbb{F}$ ). The tableau in (41) illustrates this effect; the stop in the possessive suffix surfaces faithfully (41a) to avoid a word-final labialized consonant (41b).

(41) \*C<sup>w</sup>#  $\gg$  \*uC[−round] (Nuu-chah-nulth)

|                                                                               |                   |             |              |                 |
|-------------------------------------------------------------------------------|-------------------|-------------|--------------|-----------------|
|                                                                               | *C <sup>w</sup> # | *uC[−round] | IDENT(round) | *C <sup>w</sup> |
| /meʔitḥqats-ʔi s-uk/<br>1 2 3 4 5 6 7 8 9 10 11 12 13                         |                   |             |              |                 |
| ☞ a. [.mɛ.ʔitḥ.qats-ʔi s.suk <sup>h</sup> .]<br>1 2 3 4 5 6 7 8 9 10 11 12 13 |                   | 1           |              |                 |
| b. [.mɛ.ʔitḥ.qats-ʔi s.suk <sup>wh</sup> .]<br>1 2 3 4 5 6 7 8 9 10 11 12 13  | W 1               | L           | W 1          | W 1             |

One other ranking of note in this analysis is the ranking \*C<sup>w</sup>#  $\gg$  IDENT(round)  $\gg$  \*C<sup>w</sup>. The phonotactic constraints \*C<sup>w</sup># and \*C<sup>w</sup> are in a specific/general relationship: every candidate that violates the former also violates the latter, but not every candidate



that violates the latter violates the former. Ranking the specific constraint  $*C^w\#$  above the general constraint  $*C^w$  is what Prince and Smolensky (1993/2004:99) dub a *Pāṇinian constraint relation*, after the Sanskrit (**saṅ**; Indo-European) grammarian (see Kiparsky, 1995). Ranking a specific constraint over a general constraint is the most salient way to observe the effects of the former. Under the opposite ranking,  $*C^w \gg \text{IDENT}(\text{round}) \gg *C^w\#$ , labialized stops would not surface anywhere, and there would be no evidence that they are specifically avoided word-finally (this appears to be the case in child-directed speech – see Kess and Kess, 1986). This ranking also illustrates a phenomenon dubbed *the emergence of the unmarked*, where a structure that is generally tolerated is avoided in a specific context (McCarthy and Prince, 1994; Becker and Flack Potts, 2011). In addition to the positional effect in Nuuchahnulth, emergence of the unmarked effects often manifest as more stringent restrictions in affixes or reduplicants.

## 5 Summary

The premise of Optimality Theory is that input-output mappings involve competition between multiple possible surface forms generated from an underlying form. Competition is mediated by violable constraints that either penalize phonotactically illicit structures or unfaithful mappings that change underlying properties. Conflicts between constraint preferences are resolved in favor of the hierarchically dominant constraint.

This chapter presents an overview of what is often called *classical OT*. It is one of several architectures that adheres to the properties above. Classical OT assumes a number of hypotheses that vary across architectures, among them freedom of analysis, that GEN is unrestricted, Correspondence Theory, that candidates encode their relation to the input, Consistency of Exponence, that GEN respects morphological affiliation, Richness of the Base, that the input is unrestricted. As the chapter discusses, these hypotheses have all inspired productive literature and alternative frameworks that do without them. At their core, these architectures all share the idea that hierarchical violable constraints choose between alternative surface forms.

On a continuum between entirely input-oriented frameworks and entirely output-oriented frameworks, OT occupies the latter extreme, opposite some rule-based theories of phonology. It motivates phonological mappings teleologically, achieving more perfect surface forms while committing the fewest number of changes. This has productively motivated research to identify phonotactic generalizations and formally unify otherwise disparate processes. However, OT presents too extreme a position on the continuum as it struggles to model opaque interactions (Jensen, 1995; Kirchner, 1996; McCarthy, 1996, 2007a). For example, in Okinawan (**ryu**; Japonic), when the past tense suffix  $/-ta/$  is added to an obstruent-final root, it surfaces voiced if the root obstruent is underlyingly voiced and affricates after underlying velars (42). The underlying root-final obstruents do not surface. In a rule-based analysis, this can be modeled by first applying rules that voice or affricate the suffix stop and then applying a rule that deletes the root stop in a counterbleeding order (see Baković and Blumenfeld, [this volume](#)). This is not possible in OT in part because it is non-derivational (see McCarthy, [this volume](#) and Trommer, [this volume](#) for discussion of derivational architectures), but primarily because there is no surface motivation to change the suffix stop. All three

interactions must be motivated by phonotactic constraints on obstruent clusters, which are simultaneously satisfied by deletion. The theorem of minimal violation incorrectly predicts \*[.tu.ta.n.] as the surface form of (42a). Opacity poses a significant deficit in OT’s empirical coverage and has motivated the development of numerous formal mechanisms and alternative architectures. I raise it to highlight the problems with frameworks occupying extreme ends of continua. Focusing entirely on the output reveals new details of phonological structure, but it is equally as limiting as focusing entirely on the input. The field requires some middle path.

(42) Overapplication in Okinawan (Miyara, 2015:188)

- a. /tub-ta-n/    [.tu.da.n.]    ‘fly-PST-IND’
- b. /sak-ta-n/    [.sa.tʃa.n.]    ‘bloom-PST-IND’
- c. /tug-ta-n/    [.tu.dʒa.n.]    ‘sharpen-PST-IND’

The goal of this chapter is to provide a sufficiently detailed reference for anyone who has not seen OT before and to explain its components. I have taken care to illustrate how the framework works while cataloging its diverse implementations and highlighting some of its difficulties. There is much more to say about it than fits comfortably in a chapter, but hopefully this is a sufficiently comprehensive first step.

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